Problem Set IX, Macroeconomics III
University of Copenhagen

Department of Economics, University of Copenhagen

1 Price rigidity for some

Assume a population given by a continuum of agents of size one. The representative consumer's utility function is given by

$$U_i = C_i - \frac{1}{\phi} L_i^{\phi} \text{ with } 0 < \phi \qquad \text{(Utility function)}$$

where C_i is total consumption and L_i represents labor supply. Each consumer is subject to the following budget constraint:

$$PC_i = P_i Y_i \qquad \text{(Budget constraint)}$$

where P represents the price index in the economy, P_i and Y_i are the price and quantity of the good produced by consumer i, respectively. The production technology operated by each consumer is

$$Y_i = L_i \qquad \text{(Production function)}$$

Further, assume that the economy is characterized by monopolistic competition, giving rise to demand for good i given by

$$Y_i = \left(\frac{P_i}{P}\right)^{-\theta} Y \text{ with } \theta > 1 \qquad \text{(Demand function)}$$

where θ represents the elasticity of substitution across different goods i. Consumers choose C_i, L_i and P_i. Finally, aggregate demand is pinned down by

$$Y = \frac{M}{P}$$

where M is the money supply controlled by the central bank. Agents form rational expectation.

a) Set up the maximization problem of the representative household. Solve for the first order condition. What is the optimal price the household would like to charge, as a function of its output, the money supply and the price level? How does your result compare to a hypothetical economy in which prices are determined by perfect competition? *(Hint: you may find it useful to substitute the budget constraint and the demand and production functions into the utility function, rewriting the problem in terms of one unknown, as opposed to three.)*

b) Show that, after taking logarithms, the average desired price level equals

$$p^* = v + \gamma m + (1 - \gamma)p.$$

Small letters represent logs, such that: $\ln(X) = x$. Find an expression for γ as a function of the model parameters. Under what assumptions is γ large? Give an economic reason for your answer. *(Hint: assume that the aggregate variable p^* is the averages of the individual household variables p_i. Further, since all households are identical, $y = y_i$.)*

For the remainder of the exercise, we will suppress the constant v – you may include it in your solutions, but, for brevity, you don't need to. Now, assume that a fraction λ of households can set their prices perfectly flexibly. Their price is given by

$$p^f = \gamma m + (1 - \gamma)p.$$

The remaining households' prices are more rigid, and are set before the money supply is realized. Their prices are given by

$$p^r = \gamma \mathbb{E}[m] + (1 - \gamma)\mathbb{E}[p].$$

Thus, the overall price level in the economy is $p = \lambda p^f + (1 - \lambda)p^r$. Since this is a problem of sequential price setting (flexible price-setters set their prices last), it is possible to solve for the model's solution by backward induction.

c) Find an expression for p^f, given p^r, m and the model's parameters. In the hypothetical case of $\lambda \to 1$, what would be the price level in the economy? Explain.

d) Show that the the rigid price households set their prices such that $p^r = \mathbb{E}[m]$. What is the economic intuition behind this result?

e) Show that the equilibrium values of the price level and output are given by

$$p = \mathbb{E}[m] + \frac{\lambda\gamma}{1-(1-\gamma)\lambda}\left(m - \mathbb{E}[m]\right)$$
$$y = \frac{1-\lambda}{1-(1-\gamma)\lambda}\left(m - \mathbb{E}[m]\right)$$

Explain how the price setting influences the transmission of an unexpected change in the money supply m. Discuss the validity of the statement "only unexpected changes in the money supply affect the real economy".